

Medical Imaging Equipment Failure Diagnosis Knowledge Base

Introduction

[illegible]

inspection procedures and service considerations. The material is intended as a reference for biomedical engineers and field service personnel.

CT Scanner

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MRI Systems

MRI systems include the superconducting magnet, cryostat, helium, cryocooler, gradient coils, RF coils, gradient amplifiers, RF amplifiers, shim system, patient table and host computer. Image quality depends on magnetic field homogeneity and RF integrity. Failures include helium loss, cryocooler faults, gradient amplifier trips, RF coil defects, zipper artifacts caused by RF interference, quench events, cooling failures and table positioning problems. Regular monitoring of helium level, compressor performance, room temperature and RF shielding minimizes downtime. MRI systems include the superconducting magnet, cryostat, helium, cryocooler, gradient coils, RF coils, gradient amplifiers, RF amplifiers, shim system, patient table and host computer. Image quality depends on magnetic field homogeneity and RF integrity. Failures include helium loss, cryocooler faults, gradient amplifier trips, RF coil defects, zipper artifacts caused by RF interference, quench events, cooling failures and table positioning problems.

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General Troubleshooting

Troubleshooting begins with collecting symptoms, reviewing alarms, checking power quality, verifying environmental conditions and inspecting cables and cooling systems. Engineers should isolate mechanical, electrical, electronic and software causes before replacing parts. Calibration should follow hardware repair. Documentation of every repair improves future diagnostics and reliability analysis. Troubleshooting begins with collecting symptoms, reviewing alarms, checking power quality, verifying environmental conditions and inspecting cables and cooling systems. Engineers should isolate mechanical, electrical, electronic and software causes before replacing parts. Calibration should follow hardware repair. Documentation of every repair improves future diagnostics and reliability analysis. Troubleshooting begins with collecting symptoms, reviewing alarms, checking power quality, verifying environmental conditions and inspecting cables and cooling systems. Engineers should isolate mechanical, electrical, electronic and software causes before replacing parts. Calibration should follow hardware repair. Documentation of every repair improves future diagnostics and reliability analysis.

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Sample Questions and Answers

Q: Why are ring artifacts visible on CT images? A: Detector calibration drift or failed detector channels are common causes; perform air calibration and inspect detector modules. Q: Why did the MRI stop with a helium alarm? A: Check helium level, cryocooler operation, pressure sensors and signs of leakage. Q: Why is an ultrasound probe not detected? A: Inspect the connector, cable continuity, probe EEPROM and front-end electronics. Q: Why does the CT gantry fail to rotate? A: Investigate servo motors, encoder feedback, drive belts, interlocks and motor controllers. Q: Why is image quality noisy? A: Consider detector degradation, RF interference, cooling issues, loose cables, calibration drift or damaged electronics. Q: Why are ring artifacts visible on CT images? A: Detector calibration drift or failed detector channels are common causes; perform air calibration and inspect detector modules. Q: Why did the MRI stop with a helium alarm? A: Check helium level, cryocooler operation, pressure sensors and signs of leakage. Q: Why is an ultrasound probe not detected? A: Inspect the connector, cable continuity, probe EEPROM and front-end electronics. Q: Why does the CT gantry fail to rotate? A: Investigate servo motors, encoder feedback, drive belts, interlocks and motor controllers. Q: Why is image quality noisy? A: Consider detector degradation, RF interference, cooling issues, loose cables, calibration drift or damaged electronics. Q: Why are ring artifacts visible on CT images? A: Detector calibration drift or failed detector channels are common causes; perform air calibration and inspect detector modules. Q: Why did the MRI stop with a helium alarm? A: Check helium level, cryocooler operation, pressure sensors and signs of leakage. Q: Why is an ultrasound probe not detected? A: Inspect the connector, cable continuity, probe EEPROM and front-end electronics. Q: Why does the CT gantry fail to rotate? A: Investigate servo motors, encoder feedback, drive

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